

Referee Report, April 2010

To speak to the causal effect of income on fertility, this paper analyzes the effects of husbands' job losses on fertility. The author uses data from the 1968 to 1992 waves of the Panel Study of Income Dynamics to estimate the effects for different birth parities within a hazard model framework. Table 3 shows the main results. The author finds that husbands' displacements delay first and third births with the effects persisting to 6+ years following the displacement. The author does not find any evidence that husband's displacements affect the timing of second births, however. The author performs three robustness checks: (1) conducting the analysis only using women who have a displaced husband, (2) conducting the analysis only using women with husbands' working in manufacturing, and (3) conducting the analysis after having randomly assigned "fake" displacement dummies.

My main concerns and comments are as follows:

1. What is the identifying assumption for the empirical model?

The author should be explicit that the unbiasedness of the estimates relies on job displacements being exogenous. The author should then spend a great deal of time addressing the next question.

2. Are husbands' job displacements exogenous?

The persuasiveness of the paper rests on this question more than any other. It should be addressed in the introduction and then again throughout the paper. In particular, the sample means by displacement status (Table 2) suggest that they are *not* exogenous—they are more likely for lower socioeconomic status households. I would be interested in seeing p-values from tests that the means are the same. The author should discuss the direction of the likely bias that will arise from these systematic differences.

3. The Additional Contribution

I agree with the author that there are some appealing reasons to estimate the effects in a framework that explicitly treats the data as survival-time data, unlike the approach in Lindo (forthcoming). However, the discussion (p. 5-6) reads as very one-sided. I would recommend that the author tone down this paragraph and try to be a bit more balanced in discussing the tradeoffs of each approach.

The most compelling advantages of the approach in this paper are that it accounts for right-censoring and that it can speak to different effects on different birth parities. The reader would also benefit from some discussion about what the author gains by controlling for heterogeneity via random effects.

4. "Displacement dummies are set to take a value of one only when the woman is married..."

Placing women who have divorced their displaced husbands into the control group is not appropriate. Since changes in marital status may be one of the mechanisms through which a husband's displacement affects a woman's fertility, I see no reason to do anything special with regards to marital status. Similarly, it is inappropriate to control for marital status, urban/rural status, and possibly region of residence.

Also on the topic of controls, why control for husband's age but not the woman's age? Why not control more flexibly (with fixed effects) for birth cohort and age?

5. What is the preferred empirical specification?

The author spends way too much time in the results section discussing whether or not differing heterogeneity assumptions (no heterogeneity (WH), only woman-specific heterogeneity (UH), only parity-specific heterogeneity (MS), and both woman and parity specific heterogeneity (UM)) matter. This distracts the reader from the paper's main question: how do job losses affect fertility? Unless the author can provide a good reason that the UH and MS results are important to show, I would recommend that the author focus only on the results that do not take heterogeneity into consideration and the results that allow for heterogeneity in the most flexible manner possible.

6. Mechanisms

The author repeatedly discusses "changes in information" as the likely mechanism through which displacements affect fertility (e.g., see p. 20, first line in section 5.2) but does the data tell us anything about how the job loss affects expectations? Is there any evidence from other studies? Do individuals realize that the income shock will be permanent?

I would recommend that the author address alternative explanations for the results (besides the impact on expected income) more thoroughly and in the introduction. Like "are job displacements exogenous?" this is a first-order issue that needs to be addressed.

7. Robustness Check Using Manufacturing Sample

I agree with the author that limiting the sample to households in which the husbands work in manufacturing should help achieve a better match between treatments and controls.

Unfortunately, it doesn't do nearly as good of a job as one would hope. In particular, the difference in incomes across the groups is just as large as the difference based on the full sample. For this reason, I do not find this robustness check especially compelling. Again, p-values would be nice to see.

More Minor Comments:

8. Estimated Effects on Husband's Earnings

The author estimates the effects of job loss on earnings using a random-effects regression (pg. 17-18 and Figures 1 and 2). The standard approach in the literature, since Jacobsen, LaLonde and Sullivan (1993), is to use individual fixed effects regressions and possibly to use models with individual trends. For a more recent example, see Couch and Placzek (2010).

9. Weights

The author does not use any weights in the analysis. Since the original PSID sample was not a random sample of the population, this could be a problem.

10. Why are there effects on first and third births but not second births?

This is a strange finding that merits some discussion. What type of household model would be consistent with this finding?

11. Theoretical Framework

I find it surprising that the author does not separately consider child quantity and child quality in the theoretical framework. The simpler formulation that is presented may imply that "the demand for children increases with the increase in wealth" but models that include both quantity and quality are not so clear-cut.

12. Choice of Sample

The author does not use data following 1992 “since displacements were consistently asked only until 1993” (pg. 16). Since there have been so many waves since 1992, I would prefer some more discussion of this issue. Is there really no way to make use of data from later waves?

References

Jacobsen, LaLonde, and Sullivan. “Earnings Losses of Displaced Workers.” *American Economic Review* 83(4), 685-709. 1993.

Couch, Kenneth A. and Dana W. Placzek. “Earnings Losses of Displaced Workers Revisited.” *American Economic Review* 100(1), 572-589. 2010.